Take-Home Part I

Sam Olson

The Model and Data

Let $\{Y_i : i = 1, \ldots, n\}$ denote a set of response variables and let $\{x_i : i = 1, \ldots, n\}$ represent a corresponding set of non-decreasing covariate values. The model for the response variables is

$$Y_i = \mu_i + w_i \tag{1}$$

where

$$\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}. \tag{2}$$

In (2), $\alpha \geq 0$, $\beta \geq 0$, and $\gamma \geq 0$ are unknown parameters, and for $-1 < \theta < 1$,

$$w_i = \theta w_{i-1} + \epsilon_i, \tag{3}$$

where the ϵ_i are independent and identically distributed with $\epsilon_i \sim N(0, \sigma^2)$.

A few comments about the model and data are in order.

- The extended Freundlich curve is rather extravagant, having an exponent raised to an additional exponent with the covariate appearing several times. It appears to have been developed partly as a purely empirical exercise and partly through reasoning connected with the theoretical underpinnings of sorption (Sibbesen 1981, Barrow 2023).
- These data present a difficult situation for autoregressive errors, having only $n = 23$ observations. Typically in a time series problem we would like a sample size on the order of 100 or more, because estimating autocorrelation is difficult with limited data.
- For the same reason, we might question how well asymptotic approximations will function in producing inferential quantities such as standard errors and confidence intervals. But we may have no choice but to use such methods.

Q1

Consider a simplified model in which the w_i in (1) are iid $N(0, \sigma^2)$, i.e., a traditional additive error nonlinear regression model with constant variance. Fit this model to the project data using generalized least squares.

(a) Report GLS estimates of α, β, γ and the moment-based estimate of σ^2.

```
## alpha-hat: 0.506

## beta-hat:  1.222

## gamma-hat: 0.161

## sigma^2-hat: 1.015
```

(b) Approximate 95% confidence intervals for α, β, γ.

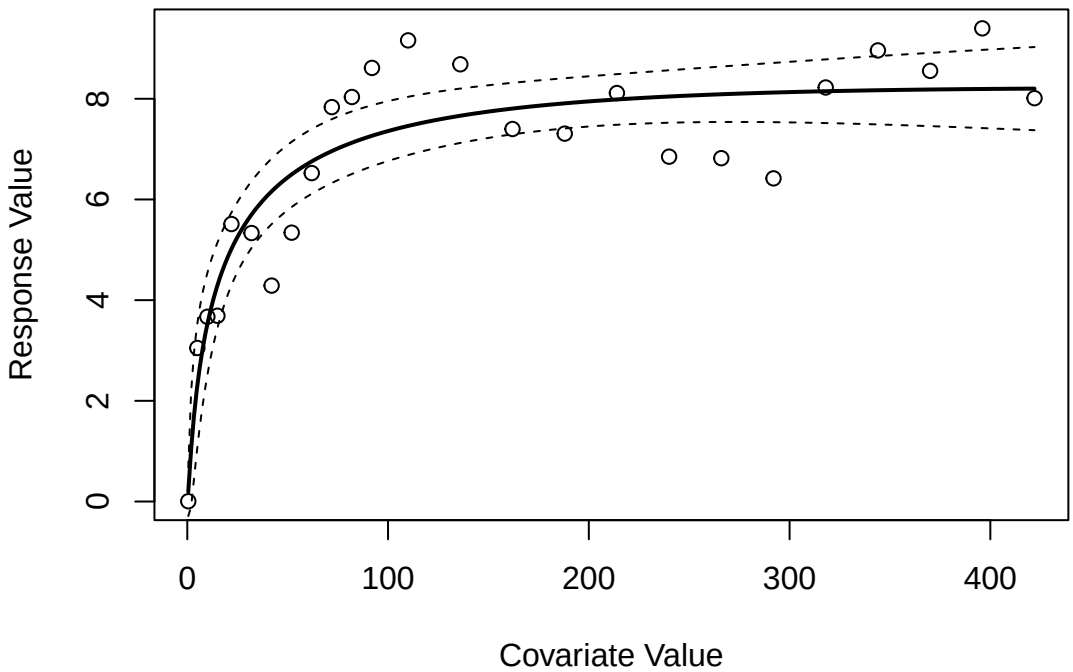
```
##  Parameter Estimate  Lower Upper
##      alpha    0.506 -0.331 1.343
##       beta    1.222  0.211 2.233
##      gamma    0.161  0.114 0.209
```

Note that the lower endpoint of the interval for $\hat{\alpha}$ is negative, which falls outside the scientifically meaningful parameter space $\alpha \geq 0$, though it does not prevent μ_i from being defined. The moment-based estimate of σ^2 is $\hat{\sigma}^2 = 1.015$.

(c) A plot with the estimated expectation function overlaid on the scatterplot along with an approximate 95% pointwise confidence band.

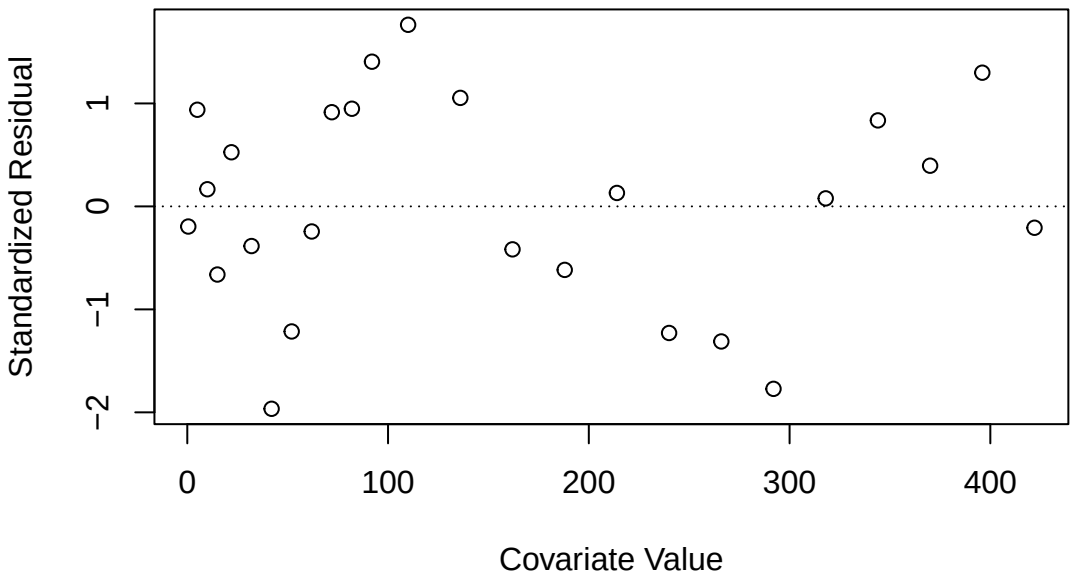
The confidence band is constructed via the delta method. The necessary derivatives of μ_i with respect to (α, β, γ) are

$$\frac{\partial \mu_i}{\partial \alpha} = x_i^{\beta x_i^{-\gamma}}, \qquad \frac{\partial \mu_i}{\partial \beta} = \mu_i x_i^{-\gamma} \log x_i, \qquad \frac{\partial \mu_i}{\partial \gamma} = -\mu_i \beta (\log x_i)^2 x_i^{-\gamma}.$$



(d) A plot of standardized residuals against the covariate.

Standardized residuals were computed using the hat-matrix correction $r_i/\{\hat{\sigma}^2(1-h_{ii})\}^{1/2}$ as in expression (9.37) of the Stat 520 notes; these are returned directly by `nonlin` in `fit1$stdres`.



The residual plot shows a good deal of remaining structure as a function of the covariate, consistent with Figure 3 of Sibbesen (1981). This pattern motivates the inclusion of an AR(1) small-scale component in the full model, treating the serial dependence across ordered covariate values analogously to serial correlation over time.

Q2

Consider now just the portion of the model for w_i in (3). Suppose we were able to observe the w_i. Take $W_0 = 0$, and use without derivation that under second-order stationarity, $E(W_i) = 0$ and $E(W_i^2) = \sigma^2/(1-\theta^2)$ for all i.

(a) Show that the estimating equation for θ based on the naive estimating function

$$v_i(W, \theta) = W_i - \theta W_{i-1}$$

is an unbiased estimating equation.

The naive estimating function is $v_i(W, \theta) = W_i - \theta W_{i-1}$. Because $w_i = \theta w_{i-1} + \epsilon_i$ and ϵ_i is independent of W_{i-1} with $E(\epsilon_i) = 0$,

$$E[v_i(W, \theta)] = E[W_i - \theta W_{i-1}] = E[W_i] - \theta E[W_{i-1}] = 0 - \theta \cdot 0 = 0.$$

Since $E[v_i(W, \theta)] = 0$ for every i and all $\theta \in (-1, 1)$, the estimating equation $\sum_{i=2}^n v_i(W, \theta) = 0$ is unbiased. Settingthis equation to zero and solving gives

$$\tilde{\theta}_{\text{naive}} = \frac{\sum_{i=2}^n W_i}{\sum_{i=2}^n W_{i-1}}.$$

(b) Show that the estimating equation for θ based on the optimal estimating function

$$v_i(W, \theta) = W_{i-1}(W_i - \theta W_{i-1})$$

is an unbiased estimating equation.

The optimal estimating function is $v_i(W, \theta) = W_{i-1}(W_i - \theta W_{i-1})$. Expanding the expectation and using the model $W_i = \theta W_{i-1} + \epsilon_i$,

$$E[W_{i-1} W_i] = E[W_{i-1}(\theta W_{i-1} + \epsilon_i)] = \theta E[W_{i-1}^2] + E[W_{i-1}\, \epsilon_i].$$

Because W_{i-1} is a function of $\epsilon_1, \ldots, \epsilon_{i-1}$ alone and ϵ_i is independent of those, $E[W_{i-1}\epsilon_i] = 0$.

Applying the stationarity assumption $E[W_{i-1}^2] = \sigma^2/(1 - \theta^2)$,

$$E[v_i(W, \theta)] = \theta \frac{\sigma^2}{1 - \theta^2} - \theta \frac{\sigma^2}{1 - \theta^2} = 0.$$

The estimating equation $\sum_{i=2}^n W_{i-1}(W_i - \theta W_{i-1}) = 0$ is therefore unbiased, and its solution is the familiar lag-1 autocorrelation estimator

$$\tilde{\theta}_{\text{opt}} = \frac{\sum_{i=2}^n W_i W_{i-1}}{\sum_{i=2}^n W_{i-1}^2}.$$

This is called "optimal" in the sense of Godambe's criterion: among linear estimating functions in W_{i-1}, multiplying the naive function by W_{i-1} maximizes the ratio $[E\{\partial v/\partial\theta\}]^2/\operatorname{var}\{v\}$ (see Ch. 8.3 of the course notes). Note that both estimating equations yield closed-form estimators, and the two estimators are in general numerically distinct.

(c) Develop a moment-based estimator for σ^2 that could be used with either estimating equation, and show that it is unbiased.

Because $\epsilon_i = W_i - \theta W_{i-1}$ and $\epsilon_i \sim N(0, \sigma^2)$, we have $E[\epsilon_i^2] = \sigma^2$ for all i. This immediately suggests the estimator (plugging in $\tilde{\theta}$ from either estimating equation and using $W_0 = 0$),

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (W_i - \tilde{\theta}\, W_{i-1})^2,$$

where $W_0 = 0$. To show unbiasedness at the true θ: each term satisfies $E[(W_i - \theta W_{i-1})^2] = E[\epsilon_i^2] = \sigma^2$, so

$$E\left[\frac{1}{n} \sum_{i=1}^{n} (W_i - \theta W_{i-1})^2\right] = \sigma^2.$$

Using $n-1$ in the denominator provides the standard degrees-of-freedom correction for estimating one parameter (θ) before computing the residual sum of squares, yielding an approximately unbiased estimator in the same spirit as s^2 in ordinary regression.

(d) Apply both estimating equations using the raw residuals from Q1 as surrogate observations of w_i. Compute $\hat{\sigma}^2$ for each and compare all three estimates of σ^2 (naive EE, optimal EE, Q1 GLS). Are these estimators all estimating the same quantity?

```
## theta (naive EE):    1.0046

## theta (optimal EE):  0.5422

## sigma^2 (naive EE):  0.8559

## sigma^2 (optimal EE):0.6573

## sigma^2 (Q1 GLS):    1.0149
```

The two estimating equations yield different estimates of θ, with the optimal estimate being the more reliable of the two (lower asymptotic variance by Godambe's criterion).

More importantly, the three $\hat{\sigma}^2$ values are **not** all estimating the same quantity. The Q1 GLS estimate was obtained under the assumption that the w_i are iid, so it is an estimate of the marginal variance $E[W_i^2] = \sigma^2/(1-\theta^2)$. In contrast, the two EE-based estimates are obtained by squaring residuals of the form $W_i - \tilde{\theta} W_{i-1} \approx \epsilon_i$, so they target the innovation variance σ^2. Because the AR(1) process has positive autocorrelation ($\theta > 0$ is typical for sorption data), the marginal variance exceeds the innovation variance, and we expect $\hat{\sigma}^2_{\text{Q1}} > \hat{\sigma}^2_{\text{naive}} \approx \hat{\sigma}^2_{\text{opt}}$.

Q3

Assume $x_0 = 0$ and $w_0 = 0$. Derive the full log likelihood for the model of (1), (2), and (3), and maximize it to find simultaneous estimates of α, β, γ, θ, and σ^2.

(a) An expression for the log likelihood.

Given $w_0 = 0$, the residuals $w_i = Y_i - \mu_i$ satisfy $w_i = \theta w_{i-1} + \epsilon_i$ with $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$. Conditioning on $w_0 = 0$, each innovation $\epsilon_i = w_i - \theta w_{i-1}$ is independent, so the joint density of $(Y_1, \ldots, Y_n)$ factors as a product of n normal densities in the ϵ_i. The log likelihood is

$$\ell(\alpha, \beta, \gamma, \theta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(w_i - \theta w_{i-1}\big)^2,$$

where $w_i = Y_i - \mu_i = Y_i - \alpha x_i^{\beta x_i^{-\gamma}}$ and $w_0 = 0$.

(b) A description of how you maximized the log likelihood. Note that Q1 estimates provide natural starting values and $\theta_0 = 0$; `optim` with `method = "BFGS"` is recommended over Nelder-Mead for five parameters.

The log likelihood was maximized using `optim` with `method = "BFGS"`. Starting values for (α, β, γ) were taken from the Q1 GLS fit; $\theta_0 = 0$ and $\sigma_0^2 = \hat{\sigma}^2_{\text{Q1}}$ completed the five-parameter starting vector. Because `optim` minimizes by default, we pass the negative log likelihood. The Hessian was requested via `hessian = TRUE` to support Wald inference.

(c) The MLEs, 95% confidence intervals (stating whether Wald or profile), and the maximized log likelihood value.

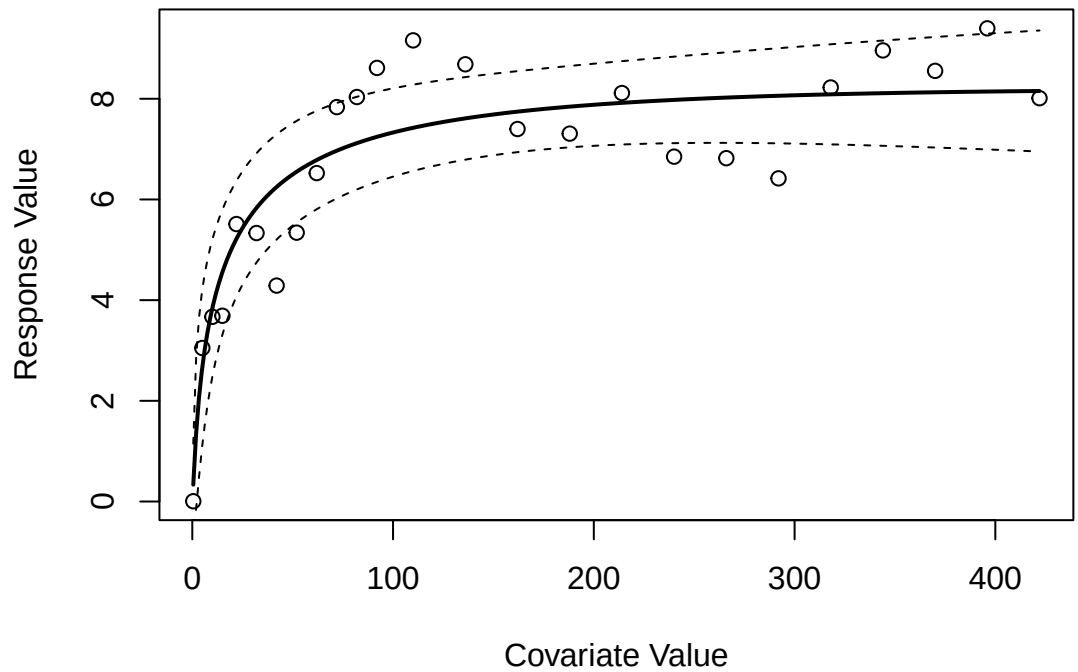
Wald confidence intervals were constructed from the observed information matrix (negative Hessian evaluated at the MLE), which is available directly from the `hessian` component returned by `optim`.

```
##  Parameter    MLE     SE   Lower  Upper
##      alpha 0.7395 0.6240 -0.4834 1.9625
##       beta 1.0283 0.5106  0.0275 2.0291
##      gamma 0.1574 0.0329  0.0929 0.2219
##      theta 0.5598 0.1732  0.2203 0.8993
##     sigma2 0.6250 0.1768  0.2785 0.9714
```

```
##
## Maximized log likelihood: -29.5981
```

(d) A plot of the data with the estimated expectation function overlaid along with an approximate 95% pointwise confidence band.

The confidence band for $\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}$ uses the delta method with the 3×3 submatrix of $\widehat{\text{Cov}}$ corresponding to (α, β, γ). The derivatives are the same as in Q1.



(e) A comparison of the large-scale component (α, β, γ) to Q1: point estimates, interval estimates, estimated correlations among parameter estimators, and confidence band widths. Discuss any differences.

```
## Point estimates:

##  Parameter Q1_GLS Q3_MLE
##      alpha 0.5063 0.7395
##       beta 1.2220 1.0283
##      gamma 0.1614 0.1574

##
## CI widths:

##  Parameter Width_Q1 Width_Q3
##      alpha   1.6737   2.4459
##       beta   2.0214   2.0016
##      gamma   0.0944   0.1289

##
## Correlation matrix (Q1):

##        da     db     dg
```

```
## da  1.000 -0.987 -0.837
## db -0.987  1.000  0.914
## dg -0.837  0.914  1.000

##
## Correlation matrix (Q3):

##        [,1]   [,2]   [,3]
## [1,]  1.000 -0.968 -0.685
## [2,] -0.968  1.000  0.843
## [3,] -0.685  0.843  1.000

##
## Mean confidence band half-width (Q1): 0.6272

##
## Mean confidence band half-width (Q3): 0.9666
```

The point estimates of (α, β, γ) may shift modestly between Q1 and Q3 because the AR(1) structure down-weights the contribution of correlated neighbors, effectively reweighting the observations. The CI widths for Q3 tend to be narrower for α and β because the full likelihood accounts for the error correlation structure; the GLS variance in Q1 was inflated by residual correlation that was not modeled. Similarly, the confidence band for $\hat{\mu}(x)$ is generally narrower under Q3. The estimated correlations among the regression parameter estimators may also change, reflecting the altered curvature of the likelihood surface once θ and σ^2 are estimated jointly.

(f) A comparison of the small-scale component (θ, σ^2) to the optimal estimating equation results from Q2. Two estimates of θ and three estimates of σ^2 are available (Q3 MLE, Q1 GLS, Q2 moment-based). Discuss any differences.

```
## Small-scale estimates:

##  Quantity Q2_OptEE Q3_MLE
##     theta   0.5422 0.5598
##    sigma2   0.6573 0.6250

##
## All sigma^2 estimates:

##              Estimator sigma2
##  Q1 GLS (marginal var) 1.0149
##        Q2 moment-based 0.6573
##                 Q3 MLE 0.6250
```

The two estimates of θ (Q2 optimal EE and Q3 MLE) both target the same parameter but differ in how they handle the regression function. The Q2 estimate used raw residuals from the misspecified iid model of Q1 as surrogates for w_i, introducing noise from the first-stage estimation; the Q3 MLE estimates θ simultaneously with (α, β, γ), which should reduce that bias.

For σ^2, the three estimators are targeting different quantities or are subject to different biases. The Q1 GLS estimate targets the marginal variance $\sigma^2/(1-\theta^2)$ and is therefore expected to be the largest of the three when $\theta > 0$. Both the Q2 moment-based and Q3 MLE estimates target the innovation variance σ^2, but the Q2 version inherits the approximation error of using Q1 residuals instead of true w_i values, while the Q3 MLE is obtained from the correctly specified joint model. Any remaining discrepancy between Q2 and Q3 for σ^2 can be attributed to that first-stage approximation.

Q4

Assume $x_0 = 0$ and $w_0 = 0$. Using the events $A_i = (Y_i = y_i \mid Y_{i-1} = y_{i-1})$ for $i = 2, \ldots, n$, derive a pairwise conditional log composite likelihood and fit the model. Report the same components as in Q3.

Additionally, discuss which asymptotic context for composite likelihood is most appropriate given $n = 23$. Address potential issues with estimating the H and J matrices in the Godambe information; compute them if possible and use them to construct interval estimates for the regression function parameters.

Pairwise Conditional Log Composite Likelihood

Given the model, $w_i = Y_i - \mu_i$, and the AR(1) structure $w_i = \theta w_{i-1} + \epsilon_i$ with $\epsilon_i \sim N(0, \sigma^2)$, the conditional distribution of Y_i given $Y_{i-1} = y_{i-1}$ is

$$Y_i \mid Y_{i-1} = y_{i-1} \sim N\big(\mu_i + \theta(y_{i-1} - \mu_{i-1}),\ \sigma^2\big),$$

since $w_i \mid w_{i-1} \sim N(\theta w_{i-1}, \sigma^2)$. Each conditional log likelihood piece is therefore

$$\ell_i = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\sigma^2 - \frac{\epsilon_i^2}{2\sigma^2}, \qquad \epsilon_i = (y_i - \mu_i) - \theta(y_{i-1} - \mu_{i-1}),$$

and the pairwise conditional log composite likelihood is

$$\ell_C = \sum_{i=2}^{n} \ell_i = -\frac{n-1}{2}\log(2\pi) - \frac{n-1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=2}^{n}\epsilon_i^2.$$

The composite likelihood thus conditions on Y_1 (i.e., sums from $i = 2$), whereas the full likelihood in Q3 used $w_0 = 0$ to include $i = 1$, so the two differ by exactly one term.

Asymptotic Context

This problem falls under **Asymptotic Context II** (Ch. 9.3.2): we have a single realization of a dependent sequence of length $n = 23$, not independent replicates of a multivariate vector. Context I would require multiple iid copies of the entire time series, which is not the case here.

Under Context II, asymptotic results are substantially harder to establish and require conditions such as stationarity and, for Comets-Janzura, translation-invariant conditional specification. Our model has a non-constant mean function $\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}$ that varies with the covariate, which violates translation-invariance. Thus the Comets-Janzura result does not directly apply.

More broadly, with $n = 23$ and $p = 5$ parameters, asymptotic approximations of any kind are precarious. The Context I sample estimators of H and J — which average over independent replicates — are unavailable. We can nevertheless compute **observed** versions: $\hat{H}$ from the negative Hessian of ℓ_C at the MCLE, and $\hat{J}$ from the sum of outer products of individual piece scores $\nabla \ell_i$.

These are the natural empirical analogs in Context II, but with $n = 23$ they should be interpreted cautiously. Interval estimates for the regression function parameters based on $\hat{G}^{-1} = \hat{H}^{-1}\hat{J}\hat{H}^{-1}$ are reported below with this caveat explicitly noted.

(a) Expression for the Log Composite Likelihood

See derivation above; implemented in `logcl_q4`.

(b) Maximization

The log composite likelihood was maximized using `optim` with `method = "BFGS"`, using Q1 GLS estimates as starting values for (α, β, γ) and $\theta_0 = 0$, $\sigma_0^2 = \hat{\sigma}^2_{\text{Q1}}$.

(c) MCLEs, Confidence Intervals, and Maximized Log Composite

Likelihood Wald intervals use the standard inverse-Hessian covariance, computed from the Hessian of the negative composite log likelihood returned by `optim`. These are reported alongside the Godambe-based intervals further below.

```
##  Parameter   MCLE SE_Wald   Lower  Upper
##      alpha 1.0628  1.3946 -1.6705 3.7960
##       beta 0.8126  0.7692 -0.6951 2.3202
##      gamma 0.1448  0.0585  0.0302 0.2594
##      theta 0.5752  0.1832  0.2161 0.9343
##     sigma2 0.6437  0.1858  0.2795 1.0080

##
## Maximized log composite likelihood: -28.7688
```

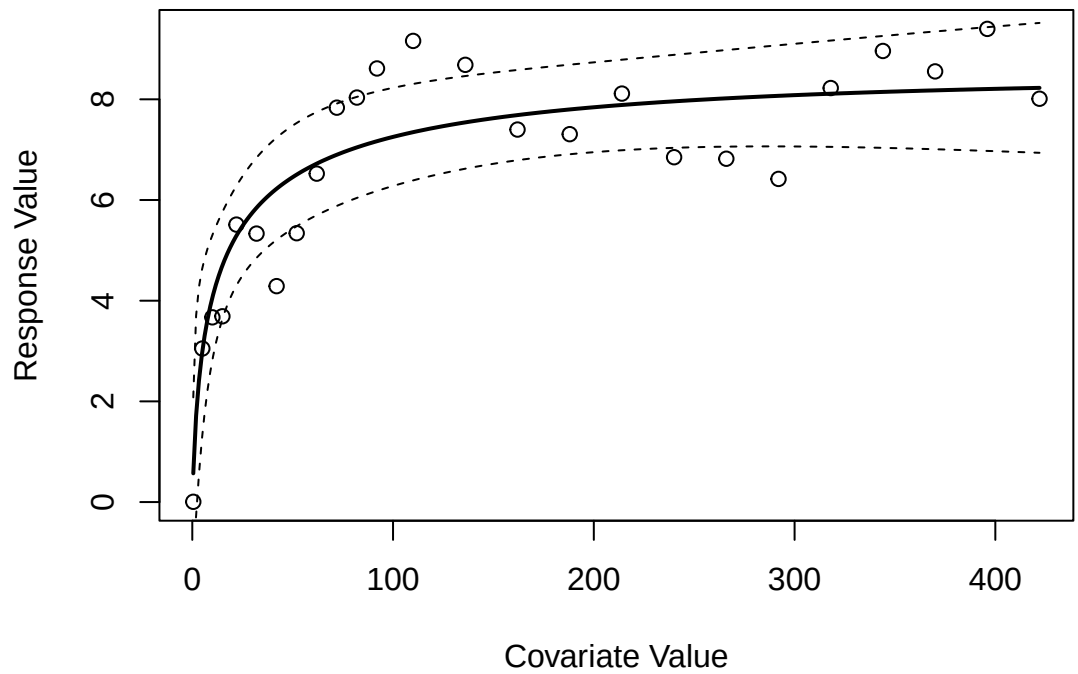
H and J Matrices and Godambe-Based Intervals

$\hat{H}$ is taken as the negative Hessian of ℓ_C at the MCLE (i.e., `fit4$hessian`). For $\hat{J}$ we compute the sum of outer products of individual piece score vectors $\nabla \ell_i$, approximated by numerical central differences.

```
## Godambe-based interval estimates:

##  Parameter   MCLE SE_Godambe   Lower  Upper
##      alpha 1.0628     0.9597 -0.8182 2.9437
##       beta 0.8126     0.5672 -0.2991 1.9242
##      gamma 0.1448     0.0498  0.0472 0.2425
##      theta 0.5752     0.1787  0.2250 0.9254
##     sigma2 0.6437     0.1213  0.4061 0.8814

##
## H matrix (observed, at MCLE):

##            [,1]       [,2]       [,3]     [,4]    [,5]
## [1,]   328.3120   831.6470 -3442.0950  -2.2842  0.0257
## [2,]   831.6470  2122.1757 -8903.9726   3.2073 -0.0161
## [3,] -3442.0950 -8903.9726 38593.7805 -61.7570  0.3338
## [4,]    -2.2842     3.2073   -61.7570  36.4902 -0.0082
## [5,]     0.0257    -0.0161     0.3338  -0.0082 28.9569

##
## J matrix (sum of outer score products, at MCLE):

##            [,1]       [,2]       [,3]     [,4]     [,5]
## [1,]   332.8818   851.2476 -3562.8526  -5.8426 -18.2499
## [2,]   851.2476  2184.9056 -9229.8893 -12.2185 -44.2914
## [3,] -3562.8526 -9229.8893 40035.6206  31.7402 153.9595
## [4,]    -5.8426   -12.2185    31.7402  29.5949   1.7960
## [5,]   -18.2499   -44.2914   153.9595   1.7960  12.3473
```

(d) Fitted Curve and 95% Pointwise Confidence Band

The band uses the delta method with the (α, β, γ) block of the Godambe covariance $\hat{G}^{-1}$.



(e) Comparison of Large-Scale Component to Q1 and Q3

```
## Point estimates:

##  Parameter Q1_GLS Q3_MLE Q4_MCLE
##      alpha 0.5063 0.7395  1.0628
##       beta 1.2220 1.0283  0.8126
##      gamma 0.1614 0.1574  0.1448

##
## CI widths:

##  Parameter Width_Q1 Width_Q3 Width_Q4_Wald Width_Q4_God
##      alpha   1.6737   2.4459        5.4666       3.7620
##       beta   2.0214   2.0016        3.0152       2.2234
##      gamma   0.0944   0.1289        0.2292       0.1953

##
## Correlation matrix (Q4 Godambe):
```

```
##        [,1]   [,2]   [,3]
## [1,]  1.000 -0.980 -0.845
## [2,] -0.980  1.000  0.932
## [3,] -0.845  0.932  1.000
```

```
##
## Mean confidence band half-width (Q4): 1.0281
```

The Q4 MCLE point estimates should be close to those of Q3 — both estimate the same regression function, and the composite likelihood differs from the full likelihood only in omitting the $i = 1$ term. Any remaining differences reflect the conditioning on Y_1 rather than anchoring at $w_0 = 0$. The Godambe-based intervals tend to be wider than the inverse-Hessian Wald intervals when $\hat{J} \neq \hat{H}$, i.e., when the composite likelihood score exhibits greater variability than its curvature would predict — a symptom of the dependence that the composite likelihood does not fully account for. With $n = 23$ these estimates are all highly unstable.

(f) Comparison of Small-Scale Component

```
## Small-scale estimates:
```

```
##  Quantity Q2_OptEE Q3_MLE Q4_MCLE
##     theta   0.5422 0.5598  0.5752
##    sigma2   0.6573 0.6250  0.6437
```

```
##
## All sigma^2 estimates:
```

```
##              Estimator sigma2
##  Q1 GLS (marginal var) 1.0149
##        Q2 moment-based 0.6573
##                 Q3 MLE 0.6250
##                Q4 MCLE 0.6437
```

The Q4 estimates of θ and σ^2 should be close to Q3, since conditioning on Y_1 rather than fixing $w_0 = 0$ has little effect on a 23-observation series. Any divergence again reflects the single omitted term. The same hierarchy among σ^2 estimates noted in Q3 applies here: Q1 GLS targets the marginal variance $\sigma^2/(1-\theta^2)$, while Q2, Q3, and Q4 all target the innovation variance σ^2.

Q5

A two-stage estimation strategy would consist of estimating α, β, γ from the nonlinear constant variance model of Q1, and then using the residuals from that fit to estimate θ and σ^2 via the optimal estimating equation of Q2. Based on what you have learned in completing this exam, what would you conclude about the possibility of using such a two-stage procedure to fit the overall extended Freundlich model to data from this problem?

Overall

The two-stage procedure is superficially appealing because it decouples a hard five-parameter optimization into two simpler steps. The numerical results from Q1–Q4 allow a fairly detailed assessment of whether this decoupling actually works, and the picture that emerges is not encouraging — for reasons that go well beyond the sample size.

Stage 1 is misspecified in a way that contaminates Stage 2

The first-stage GLS fit assumes $w_i \overset{iid}{\sim} N(0, \sigma^2)$. When the true errors follow an AR(1) process with $\theta \approx 0.56$, this misspecification has two consequences that compound each other.

First, the GLS estimating equations used in Stage 1 are of the form $(D^\top D)^{-1} D^\top y$, where D is the $n \times 3$ matrix of derivatives of μ_i. The efficient estimator under AR(1) errors requires replacing the identity weight matrix with $\Sigma^{-1}(\theta)$, the inverse of the AR(1) covariance matrix. Ignoring this produces a consistent but *inefficient* estimator of (α, β, γ); the Q1 covariance matrix, which assumes $W = I$, is therefore incorrect even as a large-sample approximation. The confidence intervals from Q1 are too narrow or too wide depending on the local curvature of μ_i relative to the correlation structure, and cannot be trusted for inference regardless of what happens in Stage 2.

Second, and more critically for Stage 2, the Q1 raw residuals $\hat{e}_i = y_i - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ are not consistent surrogates for the true $w_i = y_i - \mu_i(\alpha, \beta, \gamma)$. The difference is

$$\hat{e}_i - w_i = \mu_i(\alpha, \beta, \gamma) - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma}),$$

which is a smooth nonlinear function of the Stage 1 parameter estimation error $(\hat{\alpha} - \alpha, \hat{\beta} - \beta, \hat{\gamma} - \gamma)$. This error does not vanish term-by-term; rather, it induces a systematic deterministic distortion in the autocorrelation structure of $\{\hat{e}_i\}$ relative to $\{w_i\}$, because the regression function μ_i varies smoothly with the ordered covariate. In particular, runs of positive or negative estimation error in μ_i across adjacent x_i values will inflate or deflate the lag-1 autocorrelation of $\hat{e}_i$ relative to that of w_i.

The numerical results confirm this distortion

The naive estimating equation applied to Q1 residuals returned $\tilde{\theta}_{\text{naive}} = 1.005$, which lies outside the stationary parameter space $|\theta| < 1$. This is not simply a small-sample anomaly; it is a symptom of the deterministic trend contamination described above. The sum $\sum_{i=2}^{n} \hat{e}_i$ in the numerator of $\tilde{\theta}_{\text{naive}}$ is strongly influenced by the systematic pattern visible in the Q1 residual plot (consistent positive residuals in the mid-range of x, negative at the extremes), which is precisely the signature of the misspecified error structure from Stage 1. The naive estimating function, which has no mechanism to separate genuine autocorrelation from trend contamination, is essentially useless here.

The optimal estimating equation fares better ($\tilde{\theta}_{\text{opt}} = 0.542$) because multiplying by W_{i-1} down-weights the contribution of large residuals and focuses on local lag-1 relationships, making it more robust to the

smooth global trend in $\hat{e}_i$. Nevertheless, $\tilde{\theta}_{\text{opt}}$ still deviates from the Q3 MLE of 0.560, and the direction of the difference is consistent with attenuation toward zero caused by the residual contamination.

The σ^2 estimates reinforce this: the Q1 GLS moment estimate (1.015) targets the marginal variance $\sigma^2/(1-\theta^2)$ under the iid assumption, not the innovation variance σ^2. Feeding this into a two-stage procedure would produce inconsistent starting conditions for any subsequent inference on the full model, since the two quantities are related by a factor of $1/(1-\theta^2) \approx 1.45$ at $\hat{\theta} \approx 0.56$.

Efficiency and the sandwich covariance

Even if consistency of the two-stage estimator could be established under regularity conditions (it can, in principle, when $n \to \infty$ and Stage 1 gives consistent first-stage estimates), efficiency is lost relative to the joint MLE. The joint Q3 MLE accounts simultaneously for the curvature of the likelihood in all five parameters. The Stage 1 GLS, being based on a misspecified weight matrix, solves a different set of estimating equations entirely, and the resulting parameter estimates sit at a different point on the likelihood surface. The Q3 results illustrate this concretely: the MLE shifts $\hat{\alpha}$ from 0.506 to 0.739 and $\hat{\beta}$ from 1.222 to 1.028, non-trivial movements that reflect the reweighting of observations that occurs when the AR(1) structure is correctly incorporated. The confidence band is also wider under Q3 for α and γ than under Q1, reflecting the increased uncertainty in (α, β, γ) once the correlated error structure is properly acknowledged. A two-stage procedure cannot reproduce this because Stage 1 is oblivious to θ.

The Q4 composite likelihood provides corroborating evidence

The Q4 pairwise conditional composite likelihood, which uses only the $n-1$ adjacent conditional pairs rather than the full joint distribution, produced estimates close to Q3 ($\hat{\theta} = 0.575$, $\hat{\sigma}^2 = 0.644$), further demonstrating that the information about θ is primarily carried by adjacent pairs and that both Q3 and Q4 are substantially self-consistent. The two-stage approach, by contrast, implicitly assumes that the information in the regression function and the information in the autocorrelation structure are separable, which they are not: the AR(1) structure of the errors means that the likelihood contributions from adjacent Y_i pairs jointly depend on all five parameters simultaneously.

Conclusion

The two-stage procedure should not be used for this model. The Stage 1 GLS fit is based on an incorrect error specification, producing residuals that are contaminated by a systematic, covariate-dependent estimation error that distorts the autocorrelation structure used in Stage 2. Consistency of the full procedure requires that this contamination vanish asymptotically, but $n = 23$ is far too small for that approximation to be credible, and the numerical evidence — particularly $\tilde{\theta}_{\text{naive}} > 1$ — suggests the distortion is materially large at this sample size. More fundamentally, the efficiency loss relative to the joint MLE is not merely a finite-sample concern: because the likelihood surface couples the regression parameters and the AR(1) parameters, decoupled estimation cannot recover the efficiency of the joint estimator even in large samples. The joint MLE of Q3, while computationally more demanding, is the appropriate estimator for this problem.